

COVER PAGE

PHISHING EMAIL DETECTION & AWARENESS SYSTEM

Cyber Security Task 2 (2026)

Future Interns

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1. Executive Summary

Project Overview

The Phishing Email Detection & Awareness System is a cybersecurity project that identifies phishing emails by analyzing email headers and checking email authentication methods. It also helps users understand phishing attacks and learn how to recognize suspicious emails.

Objectives

- Detect phishing emails using email analysis.
- Analyze email headers to identify suspicious sender information.
- Verify email authenticity using SPF, DKIM, and DMARC.
- Identify phishing indicators such as fake domains and malicious links.
- Increase user awareness about phishing attacks and email security.

Tools Used

- Google Admin Toolbox – Analyzes email headers and verifies SPF, DKIM, and DMARC records.
- MXToolbox – Checks email authentication records and mail server information.
- Browser Inspection – Examines suspicious URLs, domains, and website security.
- Email Header Analysis – Reviews sender details, routing information, and email metadata.

Final Outcome

The project analyzed a sample phishing email and identified several phishing indicators, including failed SPF, DKIM, and DMARC authentication, suspicious sender domains, and unusual email header information. Based on the analysis, the email was classified as a phishing email. The project demonstrates how technical analysis and user awareness work together to improve email security and protect users from phishing attacks.

Tools Used

The project uses several free online tools to analyze suspicious emails, verify sender authenticity, and identify phishing attempts.

1. Google Admin Toolbox

Google Admin Toolbox is a free online tool used to analyze email headers and verify email authentication records.

Purpose:

- Analyze email headers.

- Verify SPF, DKIM, and DMARC records.
- Check sender authenticity.
- Review email routing information.
- Identify spoofed or suspicious emails.

2. MXToolbox

MXToolbox is an online email diagnostic tool used to verify email authentication and mail server information.

Purpose:

- Analyze email headers.
- Validate SPF records.
- Verify DKIM signatures.
- Check DMARC policies.
- Identify authentication failures and suspicious mail servers.

3. Browser Inspection

Browser Inspection is used to safely examine suspicious links and websites without interacting with harmful content.

Purpose:

- Check domain names for suspicious or fake websites.
- Verify URLs before opening them.
- Confirm HTTPS and website security.
- Inspect HTTP response headers.
- Identify redirects and phishing pages

2. Introduction

What is Phishing?

- Phishing is a **social engineering attack** used by cybercriminals to deceive users into revealing sensitive information.
- Attackers impersonate trusted organizations such as banks, government agencies, online services, or companies.
- Phishing attacks are commonly delivered through **emails, SMS messages, phone calls, and fake websites**.
- The main objective is to steal confidential information or install malicious software.
- Phishing exploits **human trust and emotions** rather than technical vulnerabilities.
- It is one of the most common and successful cyber threats worldwide.

Types of Phishing

- **Email Phishing** – Mass phishing emails sent to a large number of users.
- **Spear Phishing** – Personalized attacks targeting specific individuals or organizations.
- **Whaling** – Phishing attacks targeting senior executives and high-profile individuals.
- **Clone Phishing** – Legitimate emails are copied and modified with malicious links or attachments.
- **Business Email Compromise (BEC)** – Attackers impersonate business executives to request money or confidential information.
- **Smishing** – Phishing attacks conducted through SMS or text messages.
- **Vishing** – Voice phishing using phone calls to obtain sensitive information.

Why Phishing is Dangerous

- Steals usernames, passwords, and login credentials.
- Causes financial fraud through unauthorized transactions.
- Leads to identity theft and misuse of personal information.
- Installs malware, ransomware, spyware, and other malicious software.
- Results in data breaches and unauthorized access to confidential information.
- Causes business disruption, financial loss, and reputational damage.
- Exploits human error, making it difficult to prevent without awareness training.

Current Phishing Trends

- AI-generated phishing emails with highly convincing content.
- QR Code Phishing (**Quishing**) attacks.
- Cloud service impersonation (Microsoft 365, Google Workspace, Dropbox, etc.).
- Fake banking and online payment emails.
- Multi-factor authentication (MFA) bypass phishing attacks.
- Business Email Compromise (BEC) attacks targeting organizations.
- Mobile phishing through SMS and messaging applications.
- Fake delivery, shopping, and invoice scams.

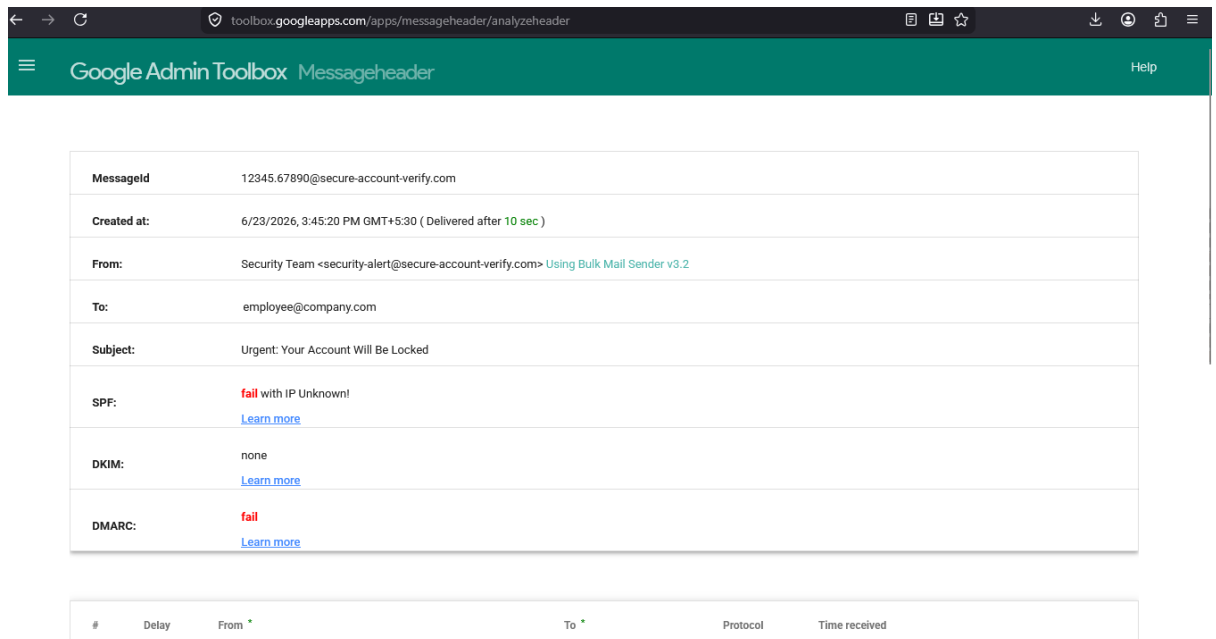
3. Objectives

The primary objective of this project is to detect phishing emails by analyzing suspicious email characteristics and identifying common phishing indicators. The project aims to examine email headers, verify email authentication protocols such as **SPF**, **DKIM**, and **DMARC**, and inspect sender information, domains, and URLs to determine whether an email is legitimate or malicious. It also focuses on understanding phishing techniques, documenting the analysis results, and creating user awareness about email security. By using tools such as **Google Admin Toolbox** and **MXToolbox**, the project demonstrates practical phishing detection methods and promotes safe email practices to help prevent credential theft, financial fraud, malware infections, and other cyber threats.

4. Tools Used

Google Admin Toolbox

Google Admin Toolbox is an online email analysis tool used to examine email headers and verify email authentication mechanisms such as **SPF**, **DKIM**, and **DMARC**. It helps identify spoofed emails and investigate phishing attempts by analyzing sender information and email routing details.



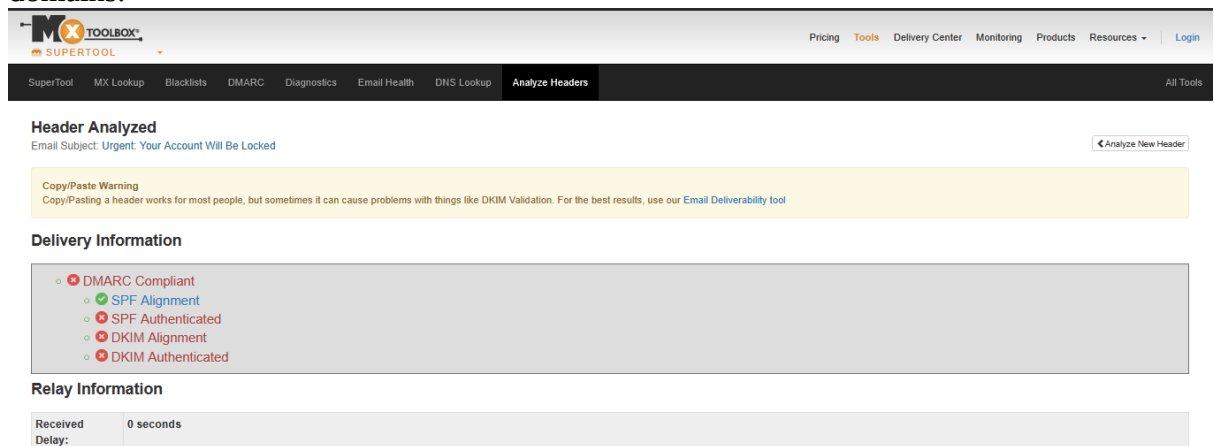
The screenshot shows the Google Admin Toolbox Messageheader interface. The header bar is green with the Google Admin Toolbox logo and a 'Messageheader' tab. Below the header, there is a table displaying email header information. The table has columns for Messageid, Created at, From, To, Subject, SPF, DKIM, and DMARC. The SPF status is 'fail with IP Unknown!' with a 'Learn more' link. The DKIM status is 'none' with a 'Learn more' link. The DMARC status is 'fail' with a 'Learn more' link. Below the table, there is a table with columns for #, Delay, From, To, Protocol, and Time received.

Messageid	Created at	From	To	Subject	SPF	DKIM	DMARC
12345.67890@secure-account-verify.com	6/23/2026, 3:45:20 PM GMT+5:30 (Delivered after 10 sec)	Security Team <security-alert@secure-account-verify.com> Using Bulk Mail Sender v3.2	employee@company.com	Urgent: Your Account Will Be Locked	fail with IP Unknown! Learn more	none Learn more	fail Learn more

#	Delay	From	To	Protocol	Time received
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MXToolbox

MXToolbox is an email diagnostic and security analysis tool used to validate email authentication records, analyze mail server information, and inspect email headers. It assists in detecting phishing emails by identifying authentication failures and suspicious sender domains.



The screenshot shows the MXToolbox Analyze Headers interface. The header bar is dark with the MXToolbox logo and a 'SUPERTOOL' dropdown. Below the header, there is a navigation bar with tabs for SuperTool, MX Lookup, Blacklists, DMARC, Diagnostics, Email Health, DNS Lookup, and Analyze Headers. The Analyze Headers tab is selected. The main content area shows the 'Header Analyzed' section with the email subject 'Urgent: Your Account Will Be Locked'. Below this, there is a 'Copy/Paste Warning' section. The 'Delivery Information' section shows a list of authentication results: DMARC Compliant, SPF Alignment, SPF Authenticated, DKIM Alignment, and DKIM Authenticated. The 'Relay Information' section shows the received delay as 0 seconds.

Header Analyzed
Email Subject: Urgent: Your Account Will Be Locked

Delivery Information
DMARC Compliant
SPF Alignment
SPF Authenticated
DKIM Alignment
DKIM Authenticated

Relay Information
Received Delay: 0 seconds

Microsoft Word / PDF

Microsoft Word was used to prepare and format the project documentation, while PDF was used to generate the final report for easy sharing, submission, and professional presentation.

GitHub

GitHub was used as a version control and project hosting platform to manage project files, documentation, screenshots, and reports. It provides a centralized repository for storing project resources and tracking updates throughout the project development process

6. Phishing Indicators

Phishing emails often contain several warning signs that help identify them as fraudulent. Recognizing these indicators is essential for preventing credential theft, malware infections, and financial fraud.

Fake Sender

Phishing emails often use fake or unfamiliar sender email addresses that appear similar to legitimate organizations. Attackers may slightly modify domain names to deceive recipients into believing the email is genuine.

Domain Spoofing

Cybercriminals create look-alike domains that closely resemble trusted company websites or email domains. These spoofed domains are designed to trick users into trusting the sender and interacting with the email.

Suspicious URLs

Phishing emails usually contain malicious or shortened URLs that redirect users to fake login pages or fraudulent websites. Users should always verify links before clicking and ensure they belong to the official domain.

Urgent Messages

Attackers frequently create a sense of urgency by using messages such as *"Your account will be suspended"* or *"Immediate action required."* This psychological tactic pressures users into acting without carefully verifying the email.

Unexpected Attachments

Many phishing emails include attachments disguised as invoices, reports, or official documents. Opening these attachments may install malware, ransomware, or spyware on the victim's device.

Grammar and Spelling Mistakes

Poor grammar, spelling errors, awkward sentence structure, and inconsistent formatting are common indicators of phishing emails. Although modern phishing attacks are becoming more sophisticated, many still contain language mistakes that reveal their fraudulent nature.

Request for Credentials

Legitimate organizations rarely ask users to provide passwords, banking details, OTPs, or other sensitive information through email. Any email requesting login credentials or confidential information should be treated as suspicious and verified through official communication channels.

7.Skills Demonstrated

Through the completion of the **Phishing Email Detection & Awareness System** project, the following technical and professional skills were demonstrated:

- **Phishing Detection** – Identified phishing emails by analyzing sender information, email content, and common phishing indicators.
-
- **Email Threat Analysis** – Performed email header analysis and verified email authentication protocols including **SPF**, **DKIM**, and **DMARC** to determine email legitimacy.
-
- **Security Awareness** – Applied cybersecurity best practices to recognize phishing attempts and developed user awareness guidelines for safe email usage.
-
- **Risk Communication** – Assessed the potential risks associated with phishing attacks, including credential theft, malware infection, financial fraud, and data breaches, and communicated the findings clearly.
-
- **Professional Documentation** – Prepared a well-structured cybersecurity project report with technical analysis, screenshots, findings, conclusions, references, and recommendations following professional documentation standards.
-
- **Email Authentication Analysis** – Interpreted authentication results from Google Admin Toolbox and MXToolbox to identify spoofed or unauthorized emails.
-
- **Cybersecurity Investigation** – Conducted systematic analysis of suspicious emails using industry-standard tools and documented the investigation process.
-
- **Analytical and Problem-Solving Skills** – Evaluated technical evidence, interpreted security findings, and classified emails based on authentication results and phishing indicators.

8.Methodology

The following methodology was followed to analyze and classify the phishing email:

Step 1: Collect Phishing Email Samples

A sample phishing email was collected for analysis. The email included the sender's address, subject, message content, and email headers required for investigation.

Step 2: Analyze Email Headers

The email header was examined to identify important fields such as **From**, **Reply-To**, **Return-Path**, **Message-ID**, and **Received**. This helped determine the origin of the email and identify any signs of spoofing or suspicious routing.

Step 3: Verify SPF, DKIM, and DMARC

The email was analyzed using **Google Admin Toolbox** and **MXToolbox** to verify the three major email authentication protocols:

- **SPF (Sender Policy Framework)**
- **DKIM (DomainKeys Identified Mail)**
- **DMARC (Domain-based Message Authentication, Reporting, and Conformance)**

The authentication results were used to determine whether the sender was authorized and whether the email had been digitally verified.

Step 4: Examine Sender, URLs, Attachments, and Domain Reputation

The sender's email address, embedded URLs, attachments, and domain reputation were inspected for phishing indicators. Suspicious domains, fake sender addresses, malicious links, and unexpected attachments were identified as potential security risks.

Step 5: Classify the Email

Based on the email header analysis, authentication results, phishing indicators, and security tool findings, the email was classified as either **Legitimate** or **Phishing**. In this project, the analyzed email failed multiple authentication checks and contained several phishing indicators; therefore, it was classified as a **High-Risk Phishing Email**.

9. Email Analysis

The collected email was analyzed using email header inspection and authentication verification to determine whether it was legitimate or a phishing attempt. The analysis focused on the sender's information, email authentication results, and common phishing indicators.

Header Analysis

The email header was examined to identify important fields such as **From**, **Reply-To**, **Return-Path**, **Message-ID**, and **Received**. The analysis revealed a suspicious sender domain, a mismatched **Reply-To** address, and the use of a bulk email sending application. These findings suggested that the email did not originate from a trusted source.

SPF, DKIM, and DMARC Results

The email authentication records were verified using **Google Admin Toolbox** and **MXToolbox**.

Authentication	Result
SPF	Failed
DKIM	None
DMARC	Failed

The failed **SPF** and **DMARC** authentication, along with the absence of a **DKIM** signature, indicated that the sender was not authorized to send emails on behalf of the claimed domain.

Sender Information

- **Sender Name:** Security Team
- **Sender Email:** security-alert@secure-account-verify.com
- **Reply-To:** support@verify-account-now.com
- **Mail Client:** Bulk Mail Sender v3.2

The sender's domain appeared suspicious and did not match a legitimate organization. The different **Reply-To** address further increased the likelihood of a phishing attempt.

Indicators of Phishing

The following phishing indicators were identified during the analysis:

- Suspicious sender email address.

- Domain spoofing using a fake domain.
- Mismatched **Reply-To** address.
- Urgent message requesting immediate account verification.
- Failed **SPF** authentication.
- Missing **DKIM** signature.
- Failed **DMARC** authentication.
- Use of bulk email sending software.
- Request to verify account credentials within a short time.

Risk Level

Risk Level: High

The email failed multiple authentication checks and contained several phishing characteristics, including spoofed sender information, authentication failures, and social engineering techniques. Based on these findings, the email was classified as a **High-Risk Phishing Email**, and users should avoid interacting with it, report it to the security team, and delete it immediately.

10. Classification

The analyzed email can be classified into one of the following categories based on the results of the email header analysis and authentication checks:

1. Safe

A **Safe** email passes email authentication checks (**SPF**, **DKIM**, and **DMARC**), originates from a trusted sender, contains no suspicious links or attachments, and does not exhibit phishing characteristics.

2. Suspicious

A **Suspicious** email shows one or more warning signs, such as an unfamiliar sender, unusual content, or partial authentication failures. These emails require additional verification before any action is taken.

3. Phishing

A **Phishing** email contains multiple indicators of a cyberattack, including fake sender information, spoofed domains, failed **SPF**, **DKIM**, or **DMARC** authentication, urgent or threatening messages, malicious links or attachments, and requests for sensitive information. Such emails should be reported immediately and deleted without interacting with them.

Final Classification

Classification: Phishing

Reason:

The analyzed email failed **SPF** and **DMARC** authentication, had no valid **DKIM** signature, used a suspicious sender domain, contained urgent language requesting account verification, and displayed multiple phishing indicators. Therefore, it is classified as a **Phishing Email** with a **High Risk** level.

11. Prevention & Awareness

Preventing phishing attacks requires both technical security measures and user awareness. By following safe email practices and recognizing common phishing indicators, users can significantly reduce the risk of becoming victims of cyberattacks.

Tips to Identify Phishing Emails

- Verify the sender's email address before trusting the message.
- Check the domain name for spelling mistakes or unusual characters.
- Be cautious of urgent or threatening messages that demand immediate action.
- Hover over links to verify the actual URL before clicking.
- Avoid opening unexpected attachments from unknown senders.
- Look for grammar, spelling, or formatting mistakes in the email.
- Never provide passwords, OTPs, or banking information through email.
- Check whether the email has passed **SPF**, **DKIM**, and **DMARC** authentication if possible.

Best Practices for Users

- Enable **Multi-Factor Authentication (MFA)** on all important accounts.
- Use strong, unique passwords and store them in a trusted password manager.
- Keep your operating system, browser, and antivirus software up to date.
- Use spam filters and email security solutions to block suspicious emails.
- Verify requests for sensitive information through official communication channels.
- Regularly participate in cybersecurity awareness and phishing training.
- Back up important data to protect against ransomware attacks.

What to Do if a Phishing Email is Received

- **Do not click** any links or buttons in the email.
- **Do not open** suspicious attachments.
- **Do not reply** to the sender or provide personal information.
- Verify the email by contacting the organization through its official website or customer support.
- Report the email to your organization's IT or cybersecurity team.

- Mark the email as **Phishing** or **Spam** in your email application.
- Delete the email after reporting it.
- If credentials have already been entered, immediately change the password, enable MFA, and notify the appropriate security team.

12. Results

The phishing email was analyzed using **email header inspection**, **Google Admin Toolbox**, and **MXToolbox**. The analysis identified multiple authentication failures and phishing indicators, confirming that the email was malicious.

Summary Table of Analyzed Email

Analysis Parameter	Result
Sender Verification	Failed
SPF Authentication	Failed
DKIM Authentication	None
DMARC Authentication	Failed
Header Analysis	Suspicious
Domain Reputation	Suspicious
Phishing Indicators	Multiple Detected
Risk Level	High
Final Classification	Phishing

Key Findings

- The sender's email address belonged to a **suspicious domain**.
- **SPF authentication failed**, indicating the sending server was not authorized.
- **No valid DKIM signature** was present to verify the email's authenticity.
- **DMARC authentication failed**, confirming that the email did not comply with the sender's authentication policy.
- The email contained **urgent language** intended to pressure the recipient into taking immediate action.
- The **Reply-To** address differed from the sender's address, indicating possible email spoofing.
- Multiple phishing indicators confirmed that the email was **malicious**.
- The email was classified as a **High-Risk Phishing Email** and should not be trusted or interacted with.

13. Conclusion

The **Phishing Email Detection & Awareness System** successfully analyzed a suspicious email using email header analysis and authentication mechanisms such as **SPF**, **DKIM**, and **DMARC**. The investigation identified multiple phishing indicators, including a suspicious sender domain, failed authentication checks, and the absence of a valid DKIM signature. Based on these findings, the email was classified as a **High-Risk Phishing Email**.

The project highlights the importance of **email authentication** in protecting users from phishing attacks. Protocols such as **SPF**, **DKIM**, and **DMARC** help verify the authenticity of email senders, prevent email spoofing, and improve the reliability of email communication. Organizations that implement these authentication mechanisms can significantly reduce the risk of phishing, fraud, and unauthorized email usage.

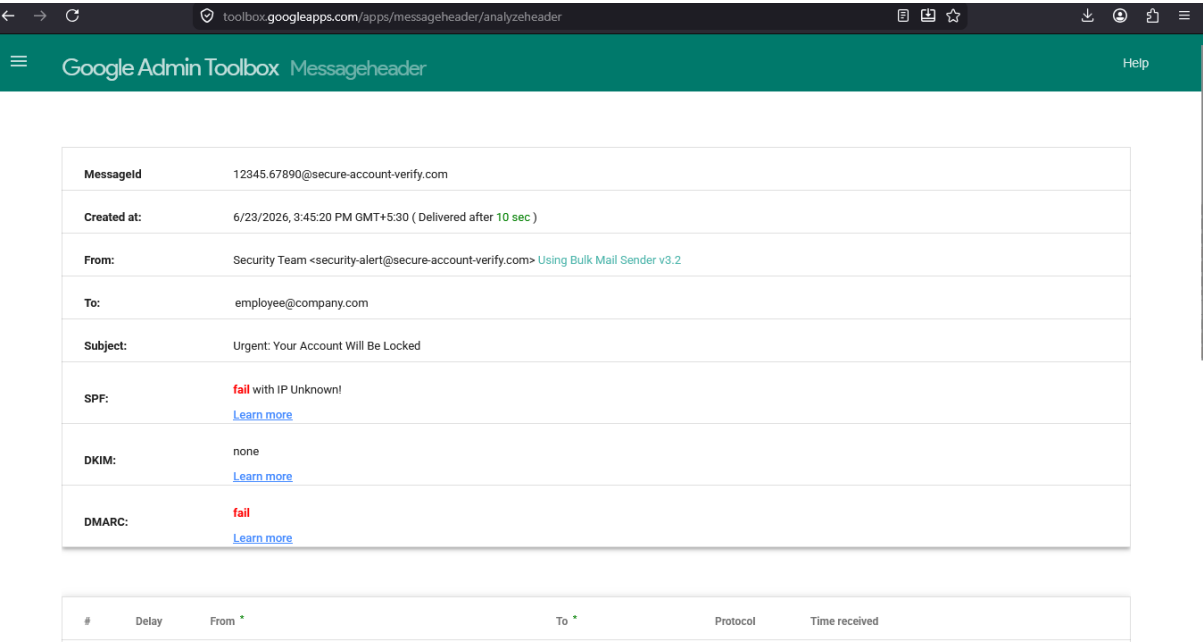
The project also emphasizes the importance of **cybersecurity awareness**. Users should always verify the sender's email address, inspect links before clicking, avoid opening unknown attachments, enable Multi-Factor Authentication (MFA), and report suspicious emails immediately. Combining strong email authentication with continuous cybersecurity awareness and safe email practices provides the most effective defense against phishing attacks and helps protect individuals and organizations from evolving cyber threats.

14. References

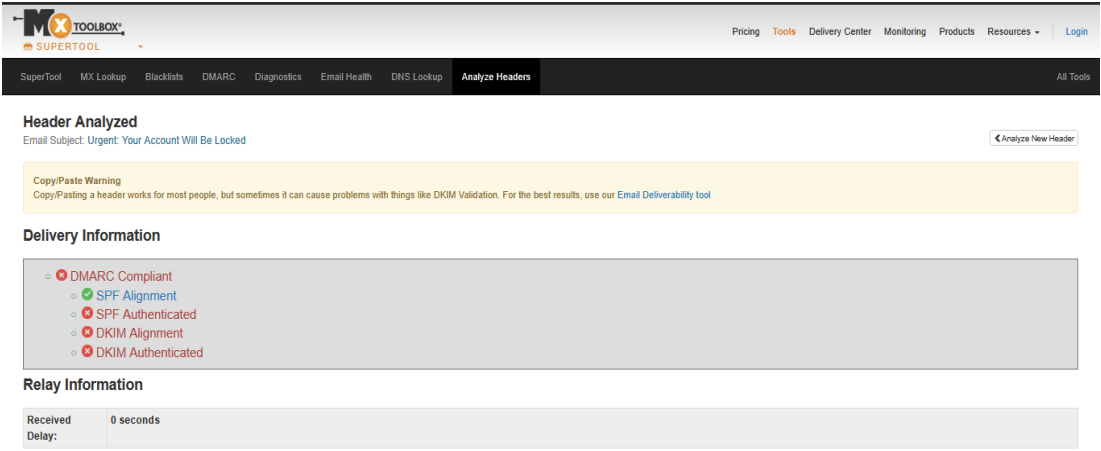
The following references were used to support the analysis and preparation of this project:

1. **Future Interns** – Cyber Security Internship Resources and Project Guidelines.
<https://futureinterns.com>
2. **Google Admin Toolbox** – Message Header Analyzer.
<https://toolbox.googleapps.com/apps/messageheader/>
3. **MXToolbox** – Email Header Analyzer and Diagnostic Tools.
<https://mxtoolbox.com>
4. **Google Workspace Documentation** – Email Authentication (SPF, DKIM, and DMARC).
<https://support.google.com/a>
5. **Microsoft Security** – Email Security and Phishing Protection.
<https://www.microsoft.com/security>
6. **OWASP (Open Worldwide Application Security Project)** – Security Awareness and Best Practices.
<https://owasp.org>
7. **RFC 7208** – *Sender Policy Framework (SPF) for Authorizing Use of Domains in Email*.
<https://www.rfc-editor.org/rfc/rfc7208>
8. **RFC 6376** – *DomainKeys Identified Mail (DKIM) Signatures*.
<https://www.rfc-editor.org/rfc/rfc6376>
9. **RFC 7489** – *Domain-based Message Authentication, Reporting, and Conformance (DMARC)*.
<https://www.rfc-editor.org/rfc/rfc7489>

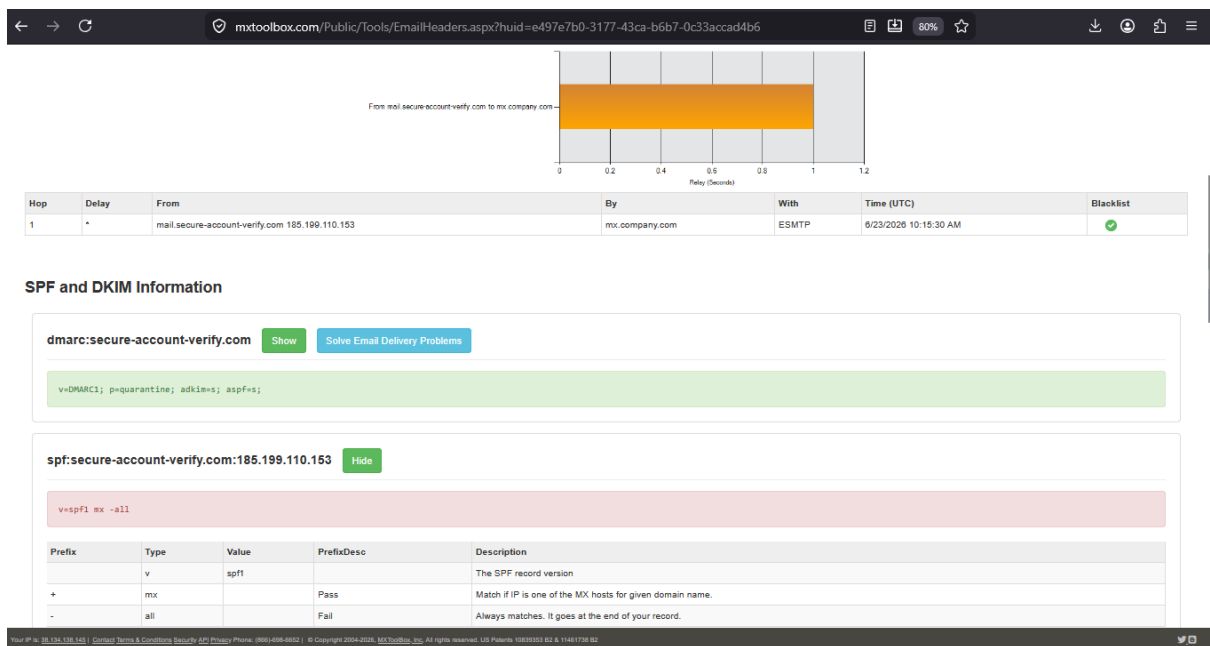
15.Appendix (Screenshots & Evidence)



Google Admin Toolbox.png



MX Toolbox.png



MX Toolbox 2.png

Test	Result	
SPF Authentication	SPF Failed for IP - 185.199.110.153	More Info
SPF Record Published	SPF Record found	
SPF Record Deprecated	No deprecated records found	
SPF Multiple Records	Less than two records found	
SPF Alignment	Domain found in SPF	
SPF Contains characters after ALL	No items after 'ALL'.	
SPF Syntax Check	The record is valid	
SPF Included Lookups	Number of included lookups is OK	
SPF Recursive Loop	Nor Recursive Loops on Includes	
SPF Duplicate Include	No Duplicate Includes Found	
SPF Type PTR Check	No type PTR found	
SPF Void Lookups	Number of void lookups is OK	
SPF MX Resource Records	Number of MX Resource Records is OK	
SPF Record Null Value	No Null DNS Lookups found	

Your DNS hosting provider is "CloudFlare" [Need Bulk Dns Provider Data?](#)

Reported by josh.ns.cloudflare.com on 6/23/2026 at 6:06:07 AM (UTC 0) [just for you](#) [Transcript](#)

Dkim Signature Error:
No DKIM-Signature header found - [more info](#)

Dkim Signature Error:
There must be at least one aligned DKIM-Signature for the message to be considered aligned. - [more info](#)

MX Toolbox 3.png

Received Header
<pre>Delivered-To: employee@company.com Received: from mail.secure-account-verify.com (185.199.110.153) by mx.company.com with ESMTP id A182C3D4 for <employee@company.com>; Tue, 23 Jun 2026 10:15:30 +0000 Received-SPF: fail (company.com: domain of security-alert@secure-account-verify.com does not designate 185.199.110.153 as permitted sender) Authentication-Results: mx.company.com; spf=fail smtp.mailfrom=security-alert@secure-account-verify.com; dkim=none; dmarc=fail Return-Path: <bounce@secure-account-verify.com> From: Security Team <security-alert@secure-account-verify.com> Reply-To: support@secure-account-verify.com To: employee@company.com Subject: Urgent: Your Account Will Be Locked Date: Tue, 23 Jun 2026 10:15:20 +0000 Message-ID: <12345.6789@secure-account-verify.com> MIME-Version: 1.0 Content-Type: text/html; charset=UTF-8 X-Mailer: Bulk Mail Sender v3.2</pre>

[Permanently forget this email header](#)

mx Toolbox 4.png

MX Toolbox

Headers Found

Header Name	Header Value
Delivered-To	employee@company.com
Received-SPF	fail (company.com: domain of security-alert@secure-account-verify.com does not designate 185.199.110.153 as permitted sender)
Authentication-Results	mx.company.com; spf=fail smtp.mailfrom=security-alert@secure-account-verify.com; dkim=none; dmarc=fail
Return-Path	<bounce@secure-account-verify.com>
From	Security Team <security-alert@secure-account-verify.com>
Reply-To	support@verify-account-now.com
To	employee@company.com
Subject	Urgent: Your Account Will Be Locked
Date	Tue, 23 Jun 2020 10:15:20 +0000
Message-ID	<12345.67890@secure-account-verify.com>
MIME-Version	1.0
Content-Type	text/html; charset=UTF-8
X-Mailer	Bulk Mail Sender v3.2

MX Toolbox 5.png